

Statics

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Act 1

We can start with the first problem, actually a good one from Morins. A damn/ A cute "Hanging Chain". In this problem a chain is attached to two points either of them on different facing off wall, and asked to find the shape of it. It is just the first part of problem. And it is actually difficult. This is the actual problem:

Problem: A chain with uniform mass density per unit length hangs between two given points on two walls. Find the general shape of the chain. Aside from an arbitrary additive constant, the function describing the shape should contain one unknown constant.

Examination: For the first time in life we will examine the solution to analyze the problem itself. This problem comes with a figure, but it is for the second part only. But a working model would work better to understand the problem for now. Now we know, that there will always be a force

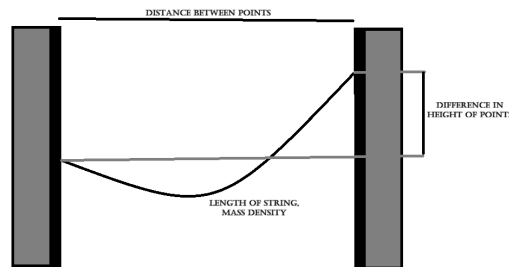


Figure 1: A hanging chain between two walls.

on the chain, do to gravity, normal force (or maybe sometime due to air drag, considerable but suspicious for now). As the chain constitute a mass, i.e. the mass density of say λ , and the total length to be l . According to the solution:

(1) Horizontal Component of chain T_x is constant throughout the chain. This exact conjecture insist us to carefully examine the situation and try to reorganise fundamentals. Through the book, it is due to the net horizontal force on any subpart of chain to be zero. Which is expected to be vague to a beginner, but not for proficient one. Breaking down this conjectural framework of solution, through careful analysis of the situation:

- We very fundamentally know that gravity is applied to this situation in general context. And as the string is attached to the walls, this string do not fall. Let's analyze it on a smaller context, let's refer to the **subparts** that were talking about. - From the figure, I can explain a set of things:

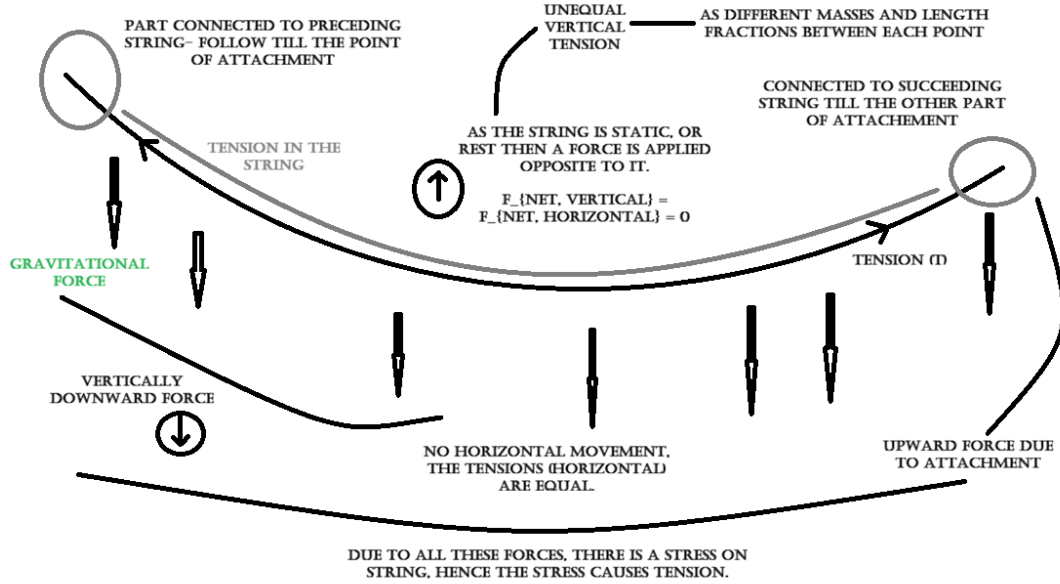


Figure 2: A hanging chain between two walls.

1) The obvious yet significant observation, is the static chain. As the chain is static both horizontally and vertically (imagine a bridge). We can basically propose a conjecture, i.e.

$$F_{net,Hori} = F_{net,Vert} = 0$$

2) As per the diagram, and solution from book, net horizontal force is constant, i.e. true conjecture. No, it is associated to law we can try to guess : **Any body at rest remain on rest until an unbalanced (apart from the net force configuration which is zero) force is applied.** Let's put a hold on our conjecture for vertical forces (assume to be $F_{net,Hori}$ not zero). **Moreover, all the horizontal tensions over the subparts are equal, as per the solution.** The simplest explanation we can basically give is, as the net force for every subpart is equal to zero, as all the subparts are static, then a constant horizontal tension can be admired. Thus stated to be $T_x = C$, for a constant "C". All our assumptions were correct. Hence connecting to static framework of book. Moving on to the problem.

-Assume the shape of chain as a function $y(x)$, from the associated problem and our knowledge of vectors, we can state:

$$\frac{T_y}{T_x} = y' \Rightarrow T_y \propto y' \quad \text{or} \quad T_y = Cy'$$

Moreover, the whole analysis of the problem is done, everything that we were supposed to learn from it are among us. Now, we will solve this problem.

The length of small piece in the image is $ds = dx\sqrt{1 + (y')^2}$. Assuming the density of chain to be constant as ρ . Then the vertical tension we have (the one we put on hold)

$$dT_y = (\rho ds)g$$

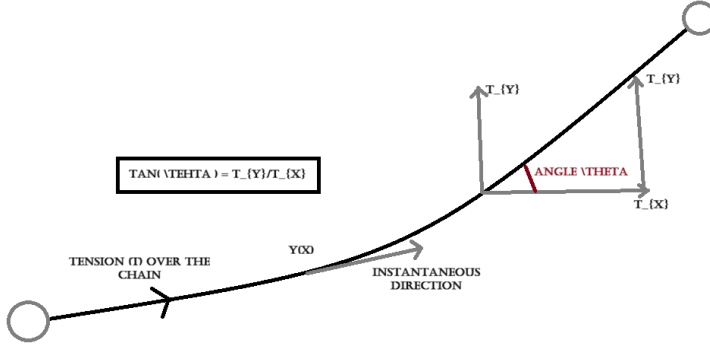


Figure 3: A hanging chain between two walls.

which is,

$$\frac{dT_y}{dx} = \rho g \sqrt{1 + (y')^2} = C y''$$

Now, we should integrate this equation till we get an answer for $y(x)$ like this:

$$y(x) = \frac{1}{\alpha} \cosh(\alpha(x + a)) + h \quad (1)$$

Where, $a = A/\alpha$, A is constant from integral for y' with $a = A/\alpha$ and $\alpha = (\rho g)/C$ being the constants. Also, h is a constant for integral of y'' . Hence the solution and so the examination is concluded. By the way I certainly introduce a few expressions that we can apply directly in any problem of this chapter.

$$\Delta F_x = \Delta F_y = \Delta \tau \text{ (net torque)} = 0 \quad (2)$$

Hence, we can get three equations conventionally, and ideally an equation that is obtained from respective constraint. These are pretty usual ways to solve problems and a few mathematical skill can also be used for harder ones. I couldn't cover these concepts in this text, but certainly the other concepts and chapter will be mentioned in respective texts in future for each necessary information.